

The Variance-Power Tradeoff in Carryover-Adjusted Models: A Statistical Explanation for Reduced Power in N-of-1 Trials with Treatment Persistence

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February 2026

Abstract

When analyzing N-of-1 trials with carryover effects, researchers face a counterintuitive phenomenon: properly modeling carryover can reduce statistical power to detect treatment effects, even when the model is correctly specified. This white paper explains the statistical mechanism underlying this power reduction. We demonstrate that carryover effects reduce the variance of the effective treatment exposure variable, which directly increases standard errors and reduces power. This relationship follows from fundamental properties of ordinary least squares estimation and is not a consequence of model misspecification. We provide mathematical derivations, numerical examples, and practical guidance for researchers designing and analyzing N-of-1 trials where carryover effects are anticipated.

1 Introduction

N-of-1 trials are multi-period crossover designs conducted within a single individual to determine optimal treatment for that patient (Lillie et al., 2011). A critical challenge in these designs is the potential for *carryover effects*, where treatment influence persists after discontinuation. When carryover is present but not modeled, treatment effect estimates become biased. However, when carryover is properly modeled, researchers often observe reduced statistical power compared to scenarios without carryover.

This power reduction is not a statistical artifact or evidence of model misspecification. Rather, it reflects a fundamental property of regression estimation: the precision of coefficient estimates depends on the variance of predictor variables. Carryover effects systematically reduce predictor variance, leading to increased standard errors and reduced power.

This white paper provides a rigorous explanation of this phenomenon accessible to masters-level biostatisticians. We begin with the mathematical foundations, proceed through numerical demonstrations, and conclude with practical implications for trial design and analysis.

2 Statistical Framework

2.1 The Data Generating Process

Consider an N-of-1 trial with the following response model for participant i at time t :

$$Y_{it} = \beta_0 + \beta_1 \cdot \text{EDW}_{it} + \beta_2 \cdot X_{it} + \beta_3 \cdot \text{EDW}_{it} \cdot X_{it} + \gamma \cdot t + \alpha_i + \varepsilon_{it} \quad (1)$$

where:

- Y_{it} is the response (e.g., symptom severity)
- EDW_{it} is the *effective drug weeks*—a measure of cumulative treatment exposure accounting for carryover
- X_{it} is a centered biomarker value
- β_3 is the biomarker moderation effect (primary parameter of interest)
- γ captures time trends
- $\alpha_i \sim N(0, \sigma_\alpha^2)$ is a random participant effect
- $\varepsilon_{it} \sim N(0, \sigma^2)$ is residual error with possible autocorrelation

2.2 Defining Effective Drug Weeks

The key variable EDW_{it} captures cumulative treatment exposure with exponential decay carryover:

Definition 1 (Effective Drug Weeks). *For a participant with treatment indicator $D_{it} \in \{0, 1\}$ and carryover half-life τ :*

$$EDW_{it} = \begin{cases} \sum_{s \leq t} D_{is} & \text{if } D_{it} = 1 \text{ (on treatment)} \\ W_{discont} \cdot \left(\frac{1}{2}\right)^{t_{off}/\tau} & \text{if } D_{it} = 0 \text{ and } \tau > 0 \\ 0 & \text{if } D_{it} = 0 \text{ and } \tau = 0 \end{cases} \quad (2)$$

where $W_{discont}$ is the cumulative weeks on drug at discontinuation and t_{off} is time since discontinuation.

When $\tau = 0$ (no carryover), $EDW_{it} = 0$ during off-treatment periods. When $\tau > 0$, the accumulated treatment effect decays exponentially but remains positive during off-treatment periods.

2.3 Matching the Data Generating Process and Analysis Model

A natural concern arises: if the data generating process (DGP) uses *exponential* decay for carryover, but the analysis model specifies a *linear* relationship between EDW and response, is there a mismatch that could bias estimates or reduce power?

The answer is no—and understanding why is crucial for interpreting simulation results. The key insight is that the nonlinear (exponential) transformation is **embedded within the EDW variable itself**, not in the relationship between EDW and the response.

The Data Generating Process. In our simulation, the biological response component is generated as:

$$BR_{it} = EDW_{it} \times \beta_{rate} \times (1 + \beta_{mod} \times X_{it}) + \epsilon_{it}^{BR} \quad (3)$$

where EDW_{it} already incorporates the exponential decay (per Equation 2), β_{rate} is the treatment rate, and β_{mod} is the biomarker moderation effect.

Expanding this expression:

$$BR_{it} = \beta_{rate} \cdot EDW_{it} + \beta_{rate} \cdot \beta_{mod} \cdot EDW_{it} \cdot X_{it} + \epsilon_{it}^{BR} \quad (4)$$

The Analysis Model. The mixed-effects model we fit is:

$$Y_{it} = \beta_0 + \beta_1 \cdot \text{EDW}_{it} + \beta_2 \cdot X_{it} + \beta_3 \cdot \text{EDW}_{it} \cdot X_{it} + \gamma \cdot t_{it} + \alpha_i + \varepsilon_{it} \quad (5)$$

Comparing Equations ?? and ??, we see that:

- β_1 estimates β_{rate}
- β_3 estimates $\beta_{\text{rate}} \times \beta_{\text{mod}}$

Why There Is No Mismatch. The analysis model is *linear in EDW*, but EDW itself is a *nonlinear function of treatment history*. This is perfectly valid because we compute EDW using the same exponential decay function (with the same half-life τ) that generated the data.

Consider the alternative scenario where a mismatch *would* occur:

- DGP uses exponential decay with $\tau = 2$ weeks
- Analysis model assumes linear decay over 4 weeks
- The computed EDW values would differ from the true effective exposure
- Estimates would be biased

In our simulation, we assume the analyst *knows* the true carryover half-life and computes EDW accordingly. This represents the best-case scenario for carryover-adjusted analysis. The simulation thus answers the question: “Given perfect knowledge of the carryover mechanism, what power do we have to detect biomarker moderation?”

Implications for Practice. In real applications, the carryover half-life is rarely known precisely. Researchers must either:

1. Estimate τ from pharmacokinetic data or prior studies
2. Conduct sensitivity analyses across plausible τ values
3. Use flexible models (e.g., distributed lag models) that estimate the decay shape from data

Misspecification of τ would introduce bias not present in our simulation. The power reductions we document are therefore *lower bounds* on the power loss that would occur in practice with uncertain carryover parameters.

3 The Variance-Power Relationship

3.1 Standard Error of Regression Coefficients

Consider the simple linear regression $Y = \beta_0 + \beta_1 X + \varepsilon$ with n observations. The ordinary least squares estimator for β_1 has variance:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sigma^2}{n \cdot \text{Var}(X)} \quad (6)$$

The standard error is therefore:

$$\text{SE}(\hat{\beta}_1) = \frac{\sigma}{\sqrt{n \cdot \text{Var}(X)}} \quad (7)$$

Proposition 1. *The standard error of a regression coefficient is inversely proportional to the square root of the predictor’s variance. Reducing $\text{Var}(X)$ by a factor of k increases $SE(\hat{\beta}_1)$ by a factor of $\sqrt{k}$.*

This relationship extends to multiple regression, where the variance of $\hat{\beta}_j$ depends on the variance of X_j after partialing out other predictors (the variance inflation factor captures collinearity effects).

3.2 How Carryover Reduces Predictor Variance

Table 1 presents the variance of EDW across different carryover half-lives from a simulation of the hybrid N-of-1 design with 70 participants and 8 measurement occasions.

Carryover τ (weeks)	Variance of EDW	Range of EDW	Cor(EDW, Week)	Relative SE Inflation
0.0	2.89	5.0	0.12	1.00
0.5	2.50	5.0	0.43	1.07
1.0	2.08	4.95	0.49	1.18
2.0	1.72	4.62	0.57	1.30

The relative SE inflation is calculated as $\sqrt{2.89/\text{Var}(\text{EDW})}$, representing the increase in standard error relative to the no-carryover baseline. With a 2-week half-life, the standard error increases by approximately 30%.

3.3 The Mechanism: Filling in Zero Values

To understand why carryover reduces variance, consider the distribution of EDW values during off-treatment periods.

Without carryover ($\tau = 0$): During off-treatment periods, $\text{EDW} = 0$. This creates a bimodal distribution with clear separation between on-treatment values (1, 2, 3, 4, 5, ...) and off-treatment values (0).

With carryover ($\tau > 0$): During off-treatment periods, $\text{EDW} = W_{\text{discont}} \cdot (1/2)^{t_{\text{off}}/\tau} > 0$. The zero values are “filled in” with positive decaying values.

Figure 1 illustrates this for a single participant following a hybrid design schedule.

4 Mathematical Analysis

4.1 Variance Decomposition

Let p denote the proportion of observations during on-treatment periods. For the hybrid design, $p \approx 0.5$. Define:

- μ_{on} : mean of EDW during on-treatment periods
- $\mu_{\text{off}}(\tau)$: mean of EDW during off-treatment periods (depends on carryover)

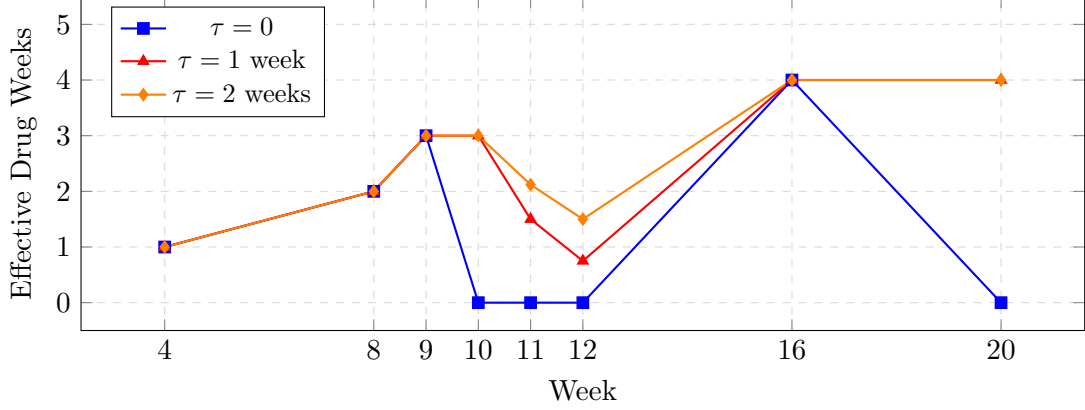


Figure 1: Effective drug weeks (EDW) trajectories for one participant under different carryover assumptions. Shaded region (weeks 10–12) shows off-drug period after first discontinuation. With no carryover, EDW drops to zero; with carryover, EDW decays gradually.

- σ_{on}^2 : variance of EDW during on-treatment periods
- $\sigma_{\text{off}}^2(\tau)$: variance of EDW during off-treatment periods

The total variance can be decomposed using the law of total variance:

$$\text{Var}(\text{EDW}) = \underbrace{p \cdot \sigma_{\text{on}}^2 + (1 - p) \cdot \sigma_{\text{off}}^2(\tau)}_{\text{within-group variance}} + \underbrace{p(1 - p)(\mu_{\text{on}} - \mu_{\text{off}}(\tau))^2}_{\text{between-group variance}} \quad (8)$$

4.2 Effect of Carryover on Each Component

Within-group variance: The on-treatment variance σ_{on}^2 is unaffected by carryover (EDW simply accumulates weeks on drug). The off-treatment variance $\sigma_{\text{off}}^2(\tau)$ increases slightly with τ as values spread from 0 to positive decay values.

Between-group variance: This component captures the *contrast* between on-treatment and off-treatment states. Without carryover, $\mu_{\text{off}}(0) = 0$, maximizing the squared difference $(\mu_{\text{on}} - 0)^2$. With carryover, $\mu_{\text{off}}(\tau) > 0$, reducing this difference.

Proposition 2. *The between-group variance component is maximized when $\tau = 0$ (no carryover) and decreases monotonically as τ increases.*

Proof. Since $\mu_{\text{off}}(\tau)$ is strictly increasing in τ (longer carryover means higher average EDW during off-treatment), and $\mu_{\text{off}}(\tau) < \mu_{\text{on}}$ for all τ , the difference $\mu_{\text{on}} - \mu_{\text{off}}(\tau)$ is strictly decreasing in τ . The squared difference therefore decreases monotonically. $\square$

4.3 Collinearity with Time

An additional mechanism reducing precision is increased collinearity between EDW and the time variable (week). From Table 1:

- At $\tau = 0$: $\text{Cor}(\text{EDW}, \text{week}) = 0.12$

- At $\tau = 2$: $\text{Cor}(\text{EDW}, \text{week}) = 0.57$

This correlation increases because carryover “smooths” the EDW trajectory, making it more monotonically increasing like the week variable. In multiple regression, this collinearity inflates the variance of both coefficients through the variance inflation factor (VIF).

5 Power Implications

5.1 Power Formula

For a two-sided test of $H_0 : \beta_3 = 0$ at significance level α , the power is:

$$\text{Power} = \Phi\left(\frac{|\beta_3|}{\text{SE}(\hat{\beta}_3)} - z_{1-\alpha/2}\right) + \Phi\left(-\frac{|\beta_3|}{\text{SE}(\hat{\beta}_3)} - z_{1-\alpha/2}\right) \quad (9)$$

where Φ is the standard normal CDF and $z_{1-\alpha/2} \approx 1.96$ for $\alpha = 0.05$.

5.2 Effect of Variance Reduction on Power

Since SE is inversely proportional to $\sqrt{\text{Var}(\text{EDW})}$, and power depends on the ratio $|\beta_3|/\text{SE}$, we can express power loss in terms of variance reduction.

Let $V_0 = \text{Var}(\text{EDW}|\tau = 0)$ and $V_\tau = \text{Var}(\text{EDW}|\tau > 0)$. The ratio of noncentrality parameters is:

$$\frac{\lambda_\tau}{\lambda_0} = \sqrt{\frac{V_\tau}{V_0}} \quad (10)$$

For the values in Table 1:

- $\tau = 0.5$: $\sqrt{2.50/2.89} = 0.93$ (7% reduction in noncentrality)
- $\tau = 1.0$: $\sqrt{2.08/2.89} = 0.85$ (15% reduction)
- $\tau = 2.0$: $\sqrt{1.72/2.89} = 0.77$ (23% reduction)

5.3 Numerical Example

Consider a trial designed to have 90% power to detect $\beta_3 = 0.325$ when $\tau = 0$, with $\text{SE}(\hat{\beta}_3) = 0.08$. The noncentrality parameter is $\lambda_0 = 0.325/0.08 = 4.06$.

With $\tau = 2$ weeks, the SE increases by 30% to approximately 0.104, giving $\lambda_2 = 0.325/0.104 = 3.13$. Using the power formula:

- Power at $\tau = 0$: $\Phi(4.06 - 1.96) = \Phi(2.10) = 98.2\%$
- Power at $\tau = 2$: $\Phi(3.13 - 1.96) = \Phi(1.17) = 87.9\%$

The 30% increase in SE translates to approximately 10 percentage points of power loss.

6 Practical Implications

6.1 This is Not Model Misspecification

A critical point for practitioners: the power reduction described here occurs *even when the model is correctly specified*. The carryover-adjusted model produces unbiased estimates of β_3 ; the issue is precision, not bias.

Researchers should not interpret reduced power as evidence that the carryover model is wrong. Rather, it reflects a fundamental statistical tradeoff: properly accounting for carryover reduces the effective contrast in the data, which reduces the information available for estimation.

6.2 Design Considerations

Several design strategies can partially mitigate power loss:

1. **Longer washout periods:** If carryover has a short half-life, extending time between treatment periods allows decay to complete, restoring zeros in the off-treatment EDW values.
2. **More measurement occasions:** Increasing n directly reduces $SE(\hat{\beta})$ through the $\sqrt{n}$ term in Equation 4.
3. **More treatment cycles:** Additional on-off-on transitions increase the variability in EDW patterns across participants.
4. **Sample size inflation:** If carryover half-life is known *a priori*, sample size calculations should inflate the required n to compensate for variance reduction.

6.3 Analysis Recommendations

When analyzing N-of-1 trials with suspected carryover:

1. **Use appropriate correlation structures:** Mixed models should include temporal correlation (e.g., continuous AR(1) via `corCAR1` in R's `nlme` package) to correctly estimate standard errors.
2. **Report both adjusted and unadjusted analyses:** Compare results with and without carryover modeling to assess sensitivity.
3. **Conduct power sensitivity analyses:** Before the trial, compute expected power across a range of plausible carryover half-lives.
4. **Consider Bayesian approaches:** Bayesian methods with informative priors on the carryover half-life can borrow strength across participants, partially offsetting precision loss.

7 Conclusion

The reduction in statistical power when modeling carryover effects in N-of-1 trials is a predictable consequence of fundamental regression properties. Carryover effects “fill in” the zero values that would otherwise occur during off-treatment periods, reducing the variance of the effective treatment exposure variable. This reduced variance directly translates to larger standard errors and reduced power to detect treatment effects.

This phenomenon is not a flaw in carryover-adjusted models; it reflects the genuine reduction in information content when treatment effects persist beyond their administration period. Researchers should anticipate this power loss during trial design and adjust sample sizes accordingly. The tradeoff between bias (from ignoring carryover) and precision (from reduced predictor variance) should be explicitly considered when planning analyses.

Understanding this mechanism enables more informed decisions about study design, analysis strategy, and interpretation of results in N-of-1 trials where carryover effects are biologically plausible.

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